

Reddy Balaji Madha

805-732-2532 | reddybalajimadha@gmail.com | [linkedin.com/in/reddy-balaji-madha](https://www.linkedin.com/in/reddy-balaji-madha) | github.com/reddybalajimadha | reddybalajimadha.github.io

EXPERIENCE

Atian.ai – Software Developer

Dec 2024 – Present

Healthcare AI Startup

Fremont, CA

- Architected containerized edge AI pipeline (**MediaMTX**, **YOLOv8n**, **Whisper**, **MedGemma 4B**) on **Ericsson Cradlepoint R1900** routers via NetCloud Manager; 1,015 transcriptions over 10hrs with zero crashes, zero cloud dependency
- Deployed 7-service edge streaming stack on ambulance Linux PCs: **Go** binaries for audio/video capture, **faster-whisper** for on-device STT, **RTSP/MediaMTX** broker, with hardware-serial-based device identity via **systemd**
- Designed reverse-SSH tunnel fleet infrastructure (**autossh** + AWS EC2 jump host) with Flask-based device registration service enabling N-edge management; reduced dead-tunnel detection from hours to 30 seconds
- Led pentest remediation; closed 2 Critical RCE vulnerabilities (**subprocess.run shell injection**), 3 High-severity findings; hardened SSH keys (17 → 6), rotated credentials, migrated to key-only auth
- Built **NEMESIS 3.5.1** XML submission pipeline in **Java/Spring Boot** with ICD-10, RxNorm, SNOMED CT mapping; multi-layer validation catching 9 wrong RxNorm + 11 wrong SNOMED codes before reaching production EHR
- Built AI guardrails for **GPT-5.2**-driven extraction: per-field code-domain validation, NEMESIS-prefix rejection, token-overlap verification against NIH RxNav; golden-test framework with 21 assertions
- Built browser-automation prototype (**Playwright** + **noVNC** + Docker) auto-filling ePCR data into ESO and Zoll EHR vendor UIs with YAML-config adapters scaling across 10+ vendor sections
- Built EMS analytics dashboard (**Flask** + **Chart.js** + **Leaflet**) covering 24 org-level KPIs; deployed to **EC2** behind **nginx** + Let's Encrypt TLS

Tata Consultancy Services (TCS) – Systems Engineer

Feb 2020 – Aug 2022

Automation & Infrastructure

Bangalore, India

- Built automation tools in **Python**, **Bash**, and **PowerShell** to streamline infrastructure operations and eliminate manual tasks
- Developed **SQL** diagnostic scripts for production incidents and built monitoring scripts across **Linux**, Windows, and macOS

TECHNICAL SKILLS

Languages: Java, Python, Go, SQL, Bash, JavaScript, TypeScript

Backend: Spring Boot, FastAPI, Flask, REST APIs, SOAP, JAXB, JWT, Playwright

AI / ML: Whisper/faster-whisper, YOLOv8, ONNX Runtime, MedGemma 4B, GPT Integration, Sentence Transformers

AI Coding Tools: Claude Code, Cursor, OpenAI Codex, GitHub Copilot

Data: Elasticsearch, PostgreSQL, Cassandra, Redis, SQLite, SQL, Data Modeling

Edge & Infra: systemd, ALSA, RTSP/MediaMTX, autossh, ffmpeg, Xvfb/noVNC, Linux (Ubuntu 24.04)

Cloud & DevOps: AWS (EC2, EKS, ECS Fargate), Docker, Kubernetes, GitHub Actions, nginx, CI/CD

Healthcare: NEMESIS 3.5.1, HL7 v2, FHIR, ICD-10, RxNorm, SNOMED CT, NIH RxNav

EDUCATION

Jessup University

Expected 2027

M.S. in Computer Science

In Progress

California Lutheran University

2022 – 2024

M.S. in Information Technology, GPA: 3.74/4.0